Geophys. J. R. astr. Soc. (1974) **36**, 267–271.

The Palaeomagnetism of some Basic Dykes Cutting the Loch Bà Felsite Ring Dyke, Isle of Mull, Scotland

J. M. Ade-Hall

(Received 1973 August 14)

Summary

The reverse natural remanent magnetism of nine basic dykes cutting the youngest unit of the Mull central intrusive complexes, the Loch Bà felsite ring dyke, is consistent with these dykes representing only part of the time interval for the intrusion of the Mull linear dyke swarm. Correlation with a second reverse polarity epoch represented in the Mull lavas and linear dyke swarm is suggested.

Introduction

The Loch Bà felsite ring dyke has a special place in the geology of the central intrusive complexes of the Palaeogene Mull volcano both as the youngest exposed intrusive unit and for its nearly perfect geometrical form (Bailey *et al.* 1924). The average diameter of the ring dyke is close to 10 km and the block of lavas and other intrusives enclosed within it can be shown to have subsided by at least 1 km. The only components of the Tertiary volcanic episode of north-west Britain known to be younger than the ring dyke are the small number of north-west trending basic dykes which cut the ring dyke. The temporal relationship of these dykes to the Mull linear dyke swarm as a whole and thus of the latter to the central intrusive complexes are questions of interest. The dykes cutting the Loch Bà felsite ring dyke are petrologically similar, and have similar NW to NNW strikes to the Mull dyke swarm, which, while focussed on the Mull intrusive centres, extends for at least 600 km from the Outer Hebrides Islands (58° 30′ N, 352° 30′ E) to Yorkshire (54° 30′ N, 359° 30′ E). For these reasons the dykes cutting the Loch Bà felsite and those of the Mull linear swarm have long been supposed to be co-eval. From radiometric dating studies it seems most probable that the whole period of this Tertiary volcanic activity was only a few million years in duration, with a median age of close to 59 My (Macintyre 1973). With the usual 5 per cent or greater magnitude of analytical and other errors involved in radiometric dating, this means that the ages of individual units of the activity cannot be resolved. The conclusion probably holds in spite of a wide range of apparent ages for volcanic material from Mull. Thus, Mussett *et al.* (1972) find ages in the 49–56 My interval, with one notable exception, from conventional whole rock K–Ar measurements for dykes from the linear swarm sampled along the north coast of Mull. However, the ring dykes and granophyres of the intrusive complexes, which are cut by only a *fraction* of the number of dykes seen in the linear swarm, yield older K–Ar ages

Downloaded from https://academic.oup.com/gji/article/36/2/267/617752 by Curtin University Library user on 26 May 2026

of close to 58 My. Again, the apparent ages of a basic dyke cutting the Loch Bà felsite ring dyke were 54·6 My by conventional K–Ar dating but 58·7 My by the argon 40–argon 39 incremental heating method (reported in Macintyre 1973). Since the latter method is likely to yield more reliable ages, the 49–56 My ages reported for the linear dyke swarm may be significantly low due to partial argon loss. In the light of this situation another approach to the problem of relating the ages of the central intrusive complexes, the dykes that cut these complexes and the linear dyke swarm is justified. Palaeomagnetism may offer such an approach. Both the Mull linear dyke swarm, sampled along the north coast of Mull, and the plateau basalts of the Gribun–Ben More area contain both normal and reverse polarities, with the sequence in both cases being: $R \rightarrow N \rightarrow R$ from oldest to youngest (Ade-Hall *et al.* 1972; Ade-Hall, Dagley & Wilson 1974). In the case of the linear dyke swarm this polarity sequence has been obtained by a comparison of dyke remanence polarities with an age sequence derived from dyke cross-cutting relationships and the chilled contact relationships of individual dykes forming multiple dyke sets. In the lavas the polarity sequence has been obtained by a comparison of remanence polarities with the stratigraphic order of flows. In both dykes and lavas normal polarity units are much less abundant than reverse polarity units, only 25 per cent of the dykes and 0·5 per cent of the lavas having normal polarity. In addition, southerly directed, shallow negative directions are commonplace. These have been termed shallow reverse directions, and are conveniently defined as reverse directions with inclinations shallower than $-55°$. One shallow reversely magnetized sheet, associated with the dykes of the linear swarm, has been shown to predate two normally magnetized dykes. Evidence for the relative ages of the central intrusive complexes and the dykes intruding them may be obtained by comparisons with the temporal sequence of directions characterizing the lava pile and the linear dyke swarm.

Sampling and magnetic measurements

The dykes sampled cut the felsite ring dyke along a ridge traverse south-west of Sròn a'Chrann-lithe on Loch Bà to a saddle 1 km south-east of the peak Beinn a'Ghraig (Fig. 1). The basic dykes have weathered down more rapidly than the felsite and *in situ* material can usually only be found in the bottom of the gulleys resulting from this process. Dyke E009 was only exposed at the bottom of a deep gorge, and since no orientation point was visible from the outcrop this dyke was not sampled. Samples were taken from dykes E001–8/10 by means of a portable diamond drill. At least four 2·5-cm diameter oriented drill cores were taken from each dyke. The natural remanent magnetization (NRM) of 2·5 cm long measurement slices was measured using a parastatic magnetometer and systematic demagnetization carried out in 5 mWb m^{-2} (50 Oe) steps up to a peak field of 20 mWb m^{-2} (200 Oe). Extra demagnetization was carried out for individual specimens where necessary to reach a final direction. As with the dykes of the linear swarm sampled along the north coast of Mull, the basic dykes cutting the felsite have very soft NRM's, implying that low deuteric oxidation titanomagnetite in excess of 10-μ diameter is the remanence carrier (Larson *et al.* 1969; Ade-Hall 1969, and many others). An unfortunate feature of the NRM of dykes E001 and 2 was the presence of very high initial moments. This feature, which is widespread in lava flows in Mull sampled from topographic highs, is probably an isothermal magnetization resulting from strong local fields associated with lightning strike. In the case of dykes with soft NRM's, it is frequently difficult to define accurately the direction of the small fraction of remanence remaining after the lightning strike magnetization has been removed. However, well defined remanence directions were obtained for six of the nine dykes sampled and polarities were clearly evident for the other three (Table 1, Fig. 2). All nine dykes show reverse remanence polarity.

Downloaded from https://academic.oup.com/gji/article/36/2/267/617752 by Curtin University Library user on 26 May 2026



FIG. 1. Location map for basic dykes cutting the Loch Bà felsite ring dyke. 1 and 10: location of dykes E001 to E010 (dykes are in numerical sequence). Outcrop of Loch Bà felsite ring dyke (part) outlined by broken line.

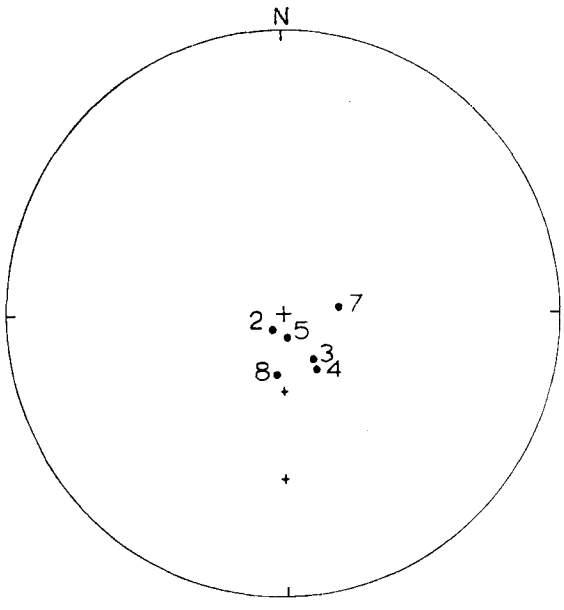


FIG. 2. Equal angle polar stereographic projection showing mean cleaned remanence directions for dykes E002/3/4/5/7/8. Filled circles imply directions in the upper hemisphere.

Downloaded from https://academic.oup.com/gji/article/36/2/267/617752 by Curtin University Library user on 26 May 2026

Table 1

Mean directions and pole positions for dykes cutting the Loch Bà felsite ring dyke, Isle of Mull

Code	Dyke thickness (m)	Portions of drill cores (m. from nearest contact)	No. collected (no. used)	Dyke mean cleaned NRM direction D(°)	I(°)	K	α_{95}(°)	Pole
A03E001	0·70	0·20, 0·31, 0·34, 0·40	4 (3)	R		105	6·6	68° N, 015° E
2	1·20	0·15, 0·43, 0·47, 0·58	4 (3)	212	−82	105	6·6	68 N, 015 E
3	0·45	0·06, 0·08, 0·12, 0·17	4 (4)	148	−69	515	2·4	71 N, 265 E
4	0·75	0·04, 0·06, 0·19, 0·24	4 (4)	151	−65	393	2·8	70 N, 247 E
5	1·20	0·19, 0·28, 0·50, 0·58	4 (4)	172	−80	117	5·1	76 N, 342 E
6	1·30	0·00, 0·08, 0·21, 0·47, 0·65		R				
7	1 to 2	0·37, 0·56, 0·70, 0·88	4 (4)	084	−68	646	2·2	38 N, 302 E
8	1·5 to 2	0·00, 0·23, 0·42, 0·52	4 (4)	186	−66	80	6·2	81 N, 148 E
9	Not sampled							
10	⩾3·5	0·60, ⩾0·85, ⩾1·20, ⩾1·40		R				
Average for E002/3/4/5/7/8			—	153	−75	33	9·9	75 N, 298 E

Downloaded from https://academic.oup.com/gji/article/36/2/267/617752 by Curtin University Library user on 26 May 2026

Discussion and conclusion

The most notable feature of the dyke remanence direction distribution is the absence of both normal or shallow reverse directions. These account for 50 per cent of the directions in the extensive Mull linear dyke swarm survey and 44 per cent in the Mull lava sequence survey and might be expected to be represented even in the small number of dykes cutting the Loch Bà felsite if the age ranges overlapped. This fact suggests that if the age duration of the basic dykes cutting the Loch Bà felsite ring dyke corresponds to any part of the direction of the Mull linear dyke swarm or of the Mull lava pile, then it must be with either the early or the late reverse polarity epochs. Since relatively very few basic dykes cut the Loch Bà felsite ring dyke it is reasonable to equate these dykes with the second, youngest, reverse polarity epoch represented elsewhere in the linear dyke swarm and the lavas.

Acknowledgments

I would like to thank Aled Evans and Alan Riding for assistance in sampling the dykes and Peter Dagley and his technical assistants for carrying out remanence measurements and computations. R. M. Macintyre was kind enough to provide the author with a preprint of his valuable summary of geochronological results for the British Tertiary Igneous province.

Department of Geology,
Dalhousie University,
Halifax,
Nova Scotia, Canada.

References

Ade-Hall, J. M., 1969. Opaque petrology and the stability of natural remanent magnetism in basaltic rocks, *Geophys. J. R. astr. Soc.*, **18,** 93–107.

Ade-Hall, J. M., Dagley, P., Wilson, R. L., Evans, A., Riding, A., Smith, P. J., Skelhorne, R. & Sloan, T., 1972. A palaeomagnetic study of the Mull regional dyke swarm, *Geophys. J. R. astr. Soc.*, **27**, 517–545.

Ade-Hall, J. M., Dagley, P. & Wilson, R. L., 1974. A palaeomagnetic study of the Mull lava sequence, *Geophys. J. R. astr. Soc.*, to be submitted.

Bailey, G. B., Clough, T. C., Wright, W B., Richey, J. E. & Wilson, G. V., 1924. Tertiary and post-Tertiary geology of Mull, Loch Aline and Oban, *Mem. geol. Surv. Scotland*, H.M.S.O.

Larson, E., Ozima, M., Ozima, M., Nagata, T. & Strangway, S., 1969. Stability of remanent magnetisation of igneous rocks, *Geophys. J. R. astr. Soc.*, **17,** 263–292.

Macintyre, R. M., 1973. Lower Tertiary geochronology of the North Atlantic continental margins. Geochronology and isotope geology of Scotland, field guide and reference. *Third European Colloquium on Geochronology*, Oxford 1973.

Mussett, A. E., Brown, G. C., Eckford, M. & Charlton, S. R., 1972. The British Tertiary Igneous Province: K–Ar ages of some dykes and lavas from Mull, Scotland, *Geophys. J. R. astr. Soc.*, **30,** 405–414.

Downloaded from https://academic.oup.com/gji/article/36/2/267/617752 by Curtin University Library user on 26 May 2026